Projects Vision of 1st Semester



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Course:
CSC-102 Programming Fundamentals

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Vision of 1st Semester Programming Fundamentals Projects

Business Application: Appointment Booking System

Vision Statement:
I want to make a console-based program that helps small businesses such as clinics, salons and tutoring centers manage their appointments. The system will allow customers to book, change and cancel appointments, and will allow the admin to manage the schedule, approve requests and view daily reports. Both users can also send messages to each other inside the program.

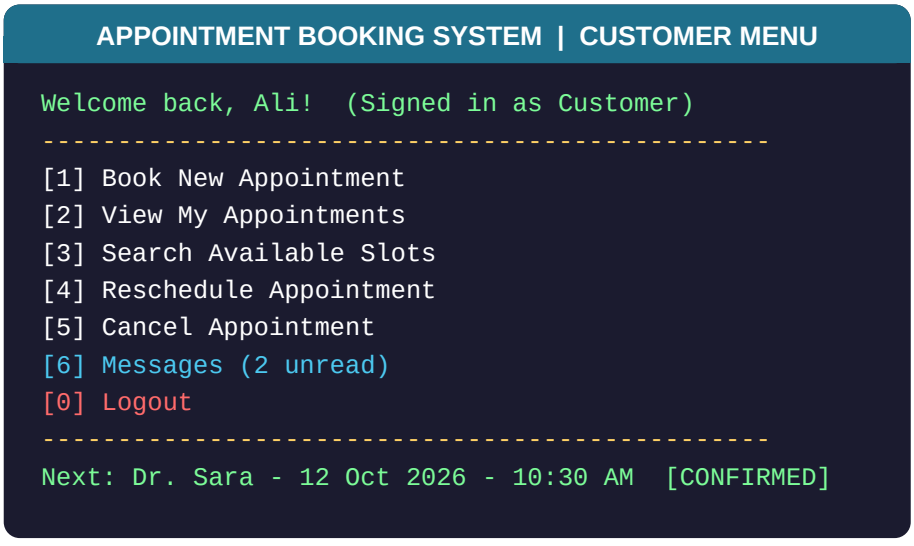
Motivation:
Many small businesses still take bookings by phone calls and write them in a paper register. This wastes time and causes problems like double bookings, forgotten appointments and long waiting for customers. I want to make a simple computer program that makes booking easier, faster and more organized.

Users:

1. **Admin (Business Owner / Manager):**
   - Add new appointment slots and services.
   - View all booked appointments.
   - Search appointments by customer name, date or ID.
   - Update, approve or reject appointments.
   - Delete appointments and view daily reports.
2. **Customer (Client):**
   - Book a new appointment.
   - View own appointments.
   - Search available slots by date or service.
   - Reschedule an appointment.
   - Cancel an appointment.

Main Features:

- Sign up and sign in for both Admin and Customer (authorized access only).
- Add, view, search, update and delete appointments.
- Messaging system between Admin and Customer (e.g. confirmation, delay or reschedule notices).
- Automatic check to prevent two customers booking the same time slot.
- Daily report of total, confirmed and cancelled appointments.
- Data saved in files so that records are not lost when the program closes.
- Colored text and properly designed headers in a simple console interface.



2D Game: Snake (Battle Edition)

Vision Statement:
I want to make a simple 2D Snake game where the player controls a snake that eats food to grow and tries to survive as long as possible. The classic game will be upgraded with three types of enemy snakes, a health and score system, multiple lives for the player, and a firing mechanism used by both the player and the enemies.

Motivation:
Snake is one of the first games I enjoyed playing on simple mobile phones. I want to turn this classic game into a more exciting one using what I learn in this course, such as loops, arrays, functions and decision making.

Gameplay Description:

- The player controls a snake that moves on a 2D grid using the keyboard.
- Eating food makes the snake grow longer and increases the score.
- Enemy snakes move around the grid and the player can fire shots to defeat them.
- The game ends when the player loses all health or all lives.

Game Features:

- **Enemy Types:**
  - Patrol Snake (slow, low health)
  - Hunter Snake (fast, chases the player, medium health)
  - Boss Serpent (high health and damage)
- **Player System:**
  - Starts with 3 lives.
  - Has a health bar.
  - Can gain an extra life after a high score.
- **Firing Mechanism:**
  - The player snake fires shots at enemies; enemy snakes spit venom at the player.
  - Hitting a wall, its own body or an enemy on contact reduces health.
- **Scoring System:**
  - +5 points for each food eaten
  - +10 points for Patrol Snake, +20 for Hunter Snake, +50 for Boss Serpent

